PID liquid filling machine — wiring reference

Arduino Nano pin assignment and full module-to-module connection table. Use this as the netlist reference when placing components in Altium (or any ECAD tool).

1. Power distribution

Main 12V/4A supply feeds two rails: raw 12V (motor driver + buck-boost input) and regulated 5V (logic and sensors, via the XL6009 buck-boost set to 5V output).

From	To	Notes
12V supply +	L298N 12V SUPPLY POSITIVE	Raw 12V for pump driver stage
12V supply +	XL6009 IN+	Input to buck-boost regulator
12V supply –	L298N GND	Common ground
12V supply –	XL6009 IN–	Common ground
XL6009 OUT+ (set to 5V)	Arduino Nano 5V pin	Powers Nano logic directly (bypasses onboard reg)
XL6009 OUT+ (set to 5V)	LCD Vcc, HX711 Vcc, Encoder Vcc	Shared regulated 5V rail
XL6009 OUT–	Arduino GND / common GND	Common ground

Important: tie all GND points together (12V supply –, L298N GND, XL6009 OUT–, Arduino GND, and every module GND) into one common ground net. Adjust the XL6009's onboard trimpot/feedback resistors to output 5.0V before connecting it to the Nano.

2. Arduino Nano pin assignment

Nano pin	Connects to	Notes
5V	Power in	From XL6009 OUT+ (5V rail)
GND	Power	Common ground
A4	I2C LCD — SDA	I2C data
A5	I2C LCD — SCL	I2C clock
D2	Rotary encoder — Clock (CLK)	Interrupt-capable pin, used for encoder quadrature
D3	Rotary encoder — Data (DT)	Interrupt-capable pin, used for encoder quadrature
D4	HX711 — Data (DOUT)	Load cell reading
D5	HX711 — SCK	Load cell clock
D6	START_BTN	To GND when pressed, use INPUT_PULLUP
D7	TARE_BTN	To GND when pressed, use INPUT_PULLUP
D8	UP_BTN	Set-point increment, use INPUT_PULLUP
D9	DOWN_BTN	Set-point decrement, use INPUT_PULLUP
D10	L298N — ENABLE	PWM output, this is the PID control signal (pump speed)
D11	L298N — IN1	Direction control
D12	L298N — IN2	Direction control

D2/D3 are deliberately used for the encoder because both support external interrupts on the Nano (ATmega328), which keeps encoder reading reliable regardless of what else the loop is doing. The 4 push buttons use the Nano's internal pull-ups, so no external resistors are needed — each button simply shorts its pin to GND when pressed.

3. Module connection detail

I2C LCD (16x2 with I2C backpack)

LCD pin	Connects to
Vcc	5V rail
GND	Common GND
SCL	Nano A5
SDA	Nano A4

Rotary encoder (KY-040)

Encoder pin	Connects to
Vcc	5V rail
GND	Common GND
Clock (CLK)	Nano D2
Data (DT)	Nano D3

Note: your pinout reference does not list an SW (push-switch) pin for the encoder. The KY-040 module typically has one; since START/TARE/UP/DOWN already have dedicated buttons, the encoder SW pin can be left unconnected.

HX711 load cell amplifier

HX711 pin	Connects to
Vcc	5V rail
GND	Common GND
Data (DOUT)	Nano D4
SCK	Nano D5

The load cell itself (4-wire: E+, E–, A+, A–) connects to the HX711's screw terminals, not directly to the Nano.

Push buttons

Button	Leg 1	Leg 2
START_BTN	Nano D6	Common GND
TARE_BTN	Nano D7	Common GND
UP_BTN	Nano D8	Common GND
DOWN_BTN	Nano D9	Common GND

L298N motor driver + pump

L298N pin	Connects to
12V SUPPLY POSITIVE	Main 12V supply +

L298N pin	Connects to
GND	Common ground
ENABLE	Nano D10 (PWM — PID output)
IN1	Nano D11
IN2	Nano D12
M1_1 MOTOR +	Pump +
M2_2 MOTOR –	Pump –

If the pump only ever runs in one direction, IN1/IN2 can be hardwired (IN1 = HIGH, IN2 = LOW) and ENABLE alone becomes the PID speed control via PWM. Wiring both to the Nano (as above) keeps the option open to reverse or brake the pump in firmware.

Buck-boost module (XL6009)

XL6009 pin	Connects to
IN+	Main 12V supply +
IN–	Main 12V supply –
OUT+	5V rail (Nano 5V, LCD, HX711, encoder Vcc)
OUT–	Common ground